

Companion XLVI: The CMB Lensing Potential in the Relaxation-Driven Cyclic Cosmology

Michael Lehmann

mi.lehmann@gmx.de

April 2026

Abstract

CMB lensing probes the integrated gravitational potential between the surface of last scattering and the observer. It encodes the geometry of spacetime, the growth of structure, and the distribution of matter across cosmic time. In the standard cosmological model, the lensing potential is determined by the sum of the two Newtonian potentials, which evolve identically in the absence of anisotropic stress. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the deflection field, the mode-coupling structure of the CMB, and the lensing power spectrum. This Companion develops a continuous description of CMB lensing in RDCC, deriving how the relaxation scale influences the lensing kernel, the deflection angle, and the reconstruction of the lensing potential. The resulting framework provides a distinctive observational signature of the RDCC gravitational field.

1 Introduction

CMB lensing arises from the deflection of cosmic microwave background photons by the gravitational potential of large-scale structure. The resulting distortions encode the integrated gravitational field along the line of sight and provide a powerful probe of the geometry and growth of structure.

In the standard cosmological model, the lensing potential is determined by the sum of the two Newtonian potentials, which evolve identically in the absence of anisotropic stress. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the lensing potential, the deflection field, and the mode-coupling structure of the CMB in a scale-dependent manner. CMB lensing becomes a direct probe of the relaxation scale k_{rel} and the temporal structure of the RDCC gravitational field.

2 The RDCC Lensing Potential

The CMB lensing potential is defined as

$$\phi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi} [\Phi + \Psi](\chi, \hat{\mathbf{n}}), \quad (1)$$

where χ_* is the comoving distance to the surface of last scattering.

In RDCC, the metric potentials evolve differently:

$$\Phi_{\text{RDCC}}(k, a) = \Phi(k, a) \left[1 - \chi_{\Phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right], \quad (2)$$

$$\Psi_{\text{RDCC}}(k, a) = \Psi(k, a) \left[1 - \chi_{\Psi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

The RDCC lensing potential becomes

$$\phi_{\text{RDCC}} = \phi \left[1 - \chi_{\phi} \left(\frac{k}{k_{\text{rel}}} \right)^{\alpha} \right]. \quad (4)$$

This modification suppresses small-scale lensing power and enhances large-scale coherence.

3 Deflection Fields and Mode Coupling

The deflection angle is given by

$$\mathbf{d}(\hat{\mathbf{n}}) = \nabla \phi(\hat{\mathbf{n}}). \quad (5)$$

In RDCC, the modified lensing potential produces a deflection field with smoother small-scale structure and enhanced large-scale coherence. The mode-coupling structure of the CMB becomes

$$\tilde{X}_{\ell m} = X_{\ell m} + \sum_{\ell' m'} \sum_{LM} \phi_{LM} \mathcal{G}_{\ell \ell' L}^{mm' M} X_{\ell' m'}, \quad (6)$$

where $\mathcal{G}$ is a geometric coupling coefficient.

Because ϕ_{LM} is modified in RDCC, the mode-coupling structure acquires a distinctive scale dependence that reflects the relaxation scale.

4 The Lensing Power Spectrum

The lensing potential power spectrum is defined as

$$C_L^{\phi\phi} = \langle \phi_{LM} \phi_{LM}^* \rangle. \quad (7)$$

In RDCC, this spectrum becomes

$$C_L^{\phi\phi, \text{RDCC}} = C_L^{\phi\phi} \left[1 - \chi_{\phi\phi} \left(\frac{L}{L_{\text{rel}}} \right)^{\alpha} \right], \quad (8)$$

where L_{rel} corresponds to the angular scale of the relaxation scale k_{rel} .

This modification suppresses small-scale lensing power and enhances large-scale structure, producing a distinctive RDCC signature.

5 Reconstruction of the Lensing Potential

CMB lensing reconstruction uses quadratic estimators to extract the lensing potential from mode coupling. The standard estimator is

$$\hat{\phi}_{LM} = N_L \sum_{\ell m} \sum_{\ell' m'} X_{\ell m} X_{\ell' m'} \mathcal{W}_{\ell \ell' L}^{mm' M}, \quad (9)$$

where N_L is a normalization factor.

In RDCC, the modified mode-coupling structure alters the reconstruction noise and the estimator response. The reconstructed potential becomes

$$\hat{\phi}_{LM, \text{RDCC}} = \hat{\phi}_{LM} \left[1 - \chi_{\text{rec}} \left(\frac{L}{L_{\text{rel}}} \right)^{\alpha} \right]. \quad (10)$$

This modification provides a direct observational probe of the relaxation scale.

6 Conclusion

CMB lensing in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the lensing potential, the deflection field, and the lensing power spectrum, producing distinctive signatures in the mode-coupling structure of the CMB. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of CMB lensing tomography, lensing reconstruction, and the interplay between matter and light in RDCC.

References

- [1] Lehmann, M. (2026). Relaxation-Driven Cyclic Cosmology (RDCC): A Minimal CPT-Symmetric Framework with a Single Parameter. Flagship Paper. Zenodo: <https://zenodo.org/records/19744958> (v43.0 or later).
- [2] The complete RDCC companion series (I–LVII) is available at the same Zenodo record above.
- [3] Companion XVI: Boltzmann Code and Modified Hierarchy.
- [4] Companion XXXI: RDCC Halo Model and Non-Linear Structure.
- [5] Companion XXXII–XXXIX: Galaxy–Halo Connection, Cosmic Web, Lensing, RSD, ISW, tSZ, Rotation Curves, CGM.
- [6] Planck Collaboration (2020), *Astron. Astrophys.* 641, A6.
- [7] DESI Collaboration (2024/2025), arXiv: [aktuelle $f\sigma_8$ /BAO-Paper].
- [8] Maiolino et al. (2024), *Nature* (JWST high- z galaxies/quasars).
- [9] Kaiser, N. (1987), *Mon. Not. R. Astron. Soc.* 227, 1 (RSD).
- [10] Bartelmann, M. & Schneider, P. (2001), *Phys. Rep.* 340, 291 (Weak Lensing Review).
- [11] Hu, W. & Okamoto, T. (2002), *Astrophys. J.* 574, 566 (CMB Lensing).
- [12] Lewis, A. & Challinor, A. (2006), *Phys. Rep.* 429, 1 (CMB Lensing Review).
- [13] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997), *Astrophys. J.* 490, 493 (Halo Profiles).